

TRIER PETRISBERG, GermanyWMO: **106090**Lat: **49.7478N**Lon: **6.6583E**Elev: **896**StdP: **14.23**Time Zone: **1.00 (EUC)**Period: **82-03**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.1	17.8	6.0	7.7	15.8	10.0	9.5	20.2	31.4	44.8	28.4	42.6	9.8	40	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	85.5	66.5	82.1	65.1	78.6	63.6	68.4	81.1	66.6	77.8	65.1	75.3	8.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.4	93.7	71.4	62.8	88.3	70.5	61.1	83.3	68.8	33.1	80.9	31.6	77.6	30.5	75.7	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.7	21.0	18.2	11.8	92.3	6.3	3.3	7.3	94.7	3.6	96.6	0.0	98.4	-4.5	100.8		
(5)			11.2	73.3	6.4	1.9	6.6	74.7	2.9	75.8	-0.7	76.9	-5.4	78.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	33.8	34.3	41.9	47.0	55.5	60.9	65.5	64.9	58.4	50.5	40.7	37.0
	DBStd	13.10	8.61	8.15	6.42	6.44	6.77	6.51	6.20	5.77	5.62	6.18	6.78	6.97
	HDD50	2125	503	440	261	131	25	2	0	0	2	69	286	406
	HDD65	5944	968	859	715	540	299	157	72	75	211	450	729	869
	CDD50	1865	0	1	10	41	196	329	482	461	253	84	6	2
	CDD65	213	0	0	0	0	6	35	88	71	12	1	0	0
	CDH74	1939	0	0	0	1	85	302	818	633	98	2	0	0
	CDH80	522	0	0	0	0	8	59	260	183	12	0	0	0
Wind	WSAvg	7.4	9.5	8.7	8.9	7.8	6.7	6.3	6.1	5.8	6.4	7.2	7.7	8.1
Precipitation	PrecAvg	29.4	2.5	2.1	2.1	1.8	2.7	2.6	3.0	2.8	2.4	2.4	2.8	2.9
	PrecMax	42.2	7.6	5.3	5.9	5.1	5.1	7.0	8.6	6.4	5.8	7.6	5.2	8.3
	PrecMin	21.3	0.9	0.3	0.4	0.0	0.5	0.3	0.3	0.6	0.6	0.2	0.1	0.1
	PrecStd	4.9	1.4	1.2	1.1	1.0	1.1	1.6	1.6	1.4	1.4	1.5	1.3	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.9	57.2	67.4	72.2	80.4	85.2	90.3	90.9	81.3	71.2	57.7	56.1
		MCWB	47.9	48.3	52.8	55.0	61.4	67.1	68.5	68.6	64.3	60.2	52.2	50.7
	2%	DB	48.7	52.2	59.3	67.0	76.1	81.5	86.4	84.6	76.8	66.1	54.8	52.2
		MCWB	45.8	46.8	48.8	52.9	60.2	65.3	67.1	65.6	63.4	57.4	51.4	49.1
	5%	DB	46.8	48.0	54.6	62.9	72.6	77.5	82.7	80.8	72.3	62.1	52.5	49.4
		MCWB	44.5	44.3	46.8	51.0	58.5	63.4	65.5	64.5	60.6	55.8	49.7	47.1
	10%	DB	44.7	45.0	51.3	58.8	68.8	73.2	78.4	76.9	68.1	59.0	50.1	46.4
		MCWB	43.0	42.3	45.8	48.5	56.7	61.8	64.2	62.6	59.4	54.4	47.7	44.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.2	50.4	53.6	56.7	64.4	70.4	70.9	69.7	66.6	61.7	54.1	51.4
		MCDB	50.7	55.1	61.9	68.7	75.5	80.9	86.1	84.4	75.5	69.0	55.6	53.1
	2%	WB	46.5	47.5	50.3	54.1	61.6	67.4	68.6	67.4	64.5	59.1	52.1	49.4
		MCDB	48.2	51.1	56.4	64.0	73.3	76.4	82.7	80.6	72.8	63.7	54.2	52.2
	5%	WB	44.7	44.8	48.3	52.0	59.5	65.3	66.7	65.7	62.6	56.7	50.0	47.1
		MCDB	46.3	47.4	53.4	60.4	70.4	73.7	78.9	77.3	69.9	60.8	52.2	49.2
	10%	WB	42.8	42.4	46.4	50.0	57.5	63.1	65.1	64.2	60.7	54.9	47.7	44.6
		MCDB	44.7	45.0	50.7	57.2	66.9	71.2	75.9	74.2	66.6	58.4	49.7	46.1
Mean Daily Temperature Range	5% DB	MDBR	7.4	10.9	13.3	16.3	17.9	17.0	19.2	19.1	16.5	12.5	8.9	7.3
		MCDBR	8.8	15.3	20.0	23.5	24.8	24.1	26.5	26.2	23.9	17.7	11.2	9.8
		MCWBR	8.0	10.8	11.3	12.3	12.7	11.1	11.3	10.9	11.5	10.1	8.9	8.6
	5% WB	MCDBR	8.8	13.8	16.6	20.3	22.5	21.2	23.6	23.0	20.3	14.8	10.2	9.4
MCWBR		8.4	10.8	10.4	11.7	12.1	10.7	10.7	10.6	10.9	9.0	8.8	8.6	
Clear-Sky Solar Irradiance	taub		0.306	0.333	0.362	0.392	0.401	0.404	0.401	0.397	0.370	0.355	0.327	0.304
	taud		2.435	2.386	2.323	2.274	2.276	2.292	2.292	2.330	2.386	2.424	2.438	2.428
	Ebn at Noon		236	256	266	270	272	272	270	266	261	244	224	219
	Edh at Noon		20	26	33	39	40	40	39	36	31	25	20	17
All-Sky Solar Radiation	RadAvg		253	483	857	1346	1590	1768	1677	1435	1076	620	285	197
	RadStd		38	91	131	168	162	158	177	147	130	59	32	29